

# Raphael Esquivel

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## Transcript of Mathematics and Physics Coursework

Lecture notes for some classes can be found at:  
<http://raffle.github.io/>.

### Harvey Mudd

#### Mathematics — Algebra

- A MATH 171**, *Abstract Algebra I*, Fall 2025, Haydee Lindo  
Groups, group actions, rings/ideals, isomorphism theorems, Sylow's theorem, representations.  
*Textbook*: Dummit & Foote (1-5, 7-9).
- A MATH 196**, *Algebraic Geometry: Scheme Theory*, Fall 2025, Dagan Karp  
Readings in graduate algebraic geometry. Introduction to the language of schemes and properties of morphisms. Sheaves, construction of schemes and morphisms between them, affine and projective schemes.  
*Textbook*: Vakil's *Rising Sea*.
- MATH 196**, *Algebraic Curves*, Fall 2026, Dagan Karp  
Readings for a second course in graduate algebraic geometry.  
*Textbook*: Eisenbud and Harris' *The Practice of Algebraic Curves*.

#### Mathematics — Analysis

- A MATH 165**, *Numerical Analysis*, Fall 2025, Jamie Haddock  
Numerical stability, nonlinear/linear solver theory, spectral decomposition algorithms, polynomial and Fourier approximation, Gaussian quadrature.  
*Textbook*: Burden & Faires' *Numerical Analysis*.

#### Mathematics — Probability & Statistics

- A- MATH 157**, *Intermediate Probability*, Spring 2026, Harrison Li  
Continuous random variables, distribution functions, joint density functions, marginal and conditional distributions, functions of random variables, conditional expectation, covariance and correlation, moment generating functions, law of large numbers, Chebyshev's theorem, and central-limit theorem.  
*Textbook*: Blitzstein & Hwang's *Introduction to Probability*.

#### Physics

- PHYS 151**, *Electromagnetic Fields*, Fall 2026, Vatche Sahakian  
The theory of static and dynamic electromagnetic fields. Multipole fields, Laplace's equation, Green's functions, the propagation of electromagnetic waves, radiation phenomena and the interaction of the electromagnetic field with matter.  
*Textbook*: None.

- A**    **PHYS 172**, *General Relativity and Cosmology*, Spring 2026, Rahulkumar Solanki  
The principle of equivalence, Riemannian geometry, and the Schwarzschild and cosmological solutions of the field equations.  
*Textbook*: None.
- A**    **PHYS 187**, *Quantum Field Theory*, Spring 2026, Kevin Setter  
Relativistic classical field theory, canonical quantization, Klein-Gordon and Dirac fields, spinors and Lorentz symmetry, interacting scalar field theory, Feynman diagrams, scattering amplitudes, quantum electrodynamics, nonabelian gauge theories, spontaneous symmetry breaking, the Higgs mechanism, elements of the Standard Model.  
*Textbook*: None.

## Other

- ENGR 079**, *Introduction to Engineering Systems*, Fall 2026, Nancy Lape  
Mathematical modeling, analysis, and control of mechanical and electrical systems, including transfer functions, Laplace transforms, transient and frequency-response analysis, Bode methods, stability, and feedback-controller design, with experimental implementation in robotic systems.  
*Textbook*: None.
- CHEM 122**, *Nanomaterials*, Fall 2026, Hal Van Ryswyk  
Experimental course in nanoscale materials synthesis, self-assembly, nanofabrication, and structure-property characterization, with work spanning quantum dots, metal nanocrystals, graphene, thin films, and photonic materials using techniques including AFM, SEM/STEM, Raman spectroscopy, UV-Vis spectroscopy, and X-ray diffraction.  
*Textbook*: None.

## Arizona State University

### Physics

- A+**    **PHY 302**, *Mathematical Methods in Physics II*, Fall 2024, Tanmay Vachaspati  
Vector calculus, complex analysis, partial differential equations, Sturm-Liouville theory, special functions, numerical methods.  
*Textbook*: None.
- A+**    **PHY 361**, *Introductory Modern Physics*, Fall 2025, Ricardo Alarcon  
Special relativity and introductory quantum theory with applications drawn from atomic, nuclear, and solid-state physics.  
*Textbook*: None.
- A**    **PHY 311**, *Classical Particles, Fields, and Matter II*, Spring 2025, Damien Easson  
Electrostatics, potential problems in 3D, boundary value problems, Poisson and Laplace equations, multipole expansion, magnetic fields and materials, magnetic induction, Faraday's Law, displacement current, Maxwell's equations.  
*Textbook*: Griffiths (1-7).

**A+**   **PHY 315**, *Quantum Physics II*, Spring 2025, Siddarth Karkare

General principles of quantum mechanics, 3-dimensional bound and scattering problems, approximation methods, time-dependent perturbation theory, spin, angular momentum, relativistic corrections, many-particle systems.

*Textbook:* Griffiths (7-12).